

Prediction error minimization, standard model of statistical machine learning, asymptotic consistency, PAC bounds

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Outline

- ▶ Stochastic state-space models as predictors.
- ▶ The standard framework for statistical learning.
- ▶ Prediction error minimization (PEM) for LTI systems.
- ▶ Asymptotic consistency of PEM for LTI systems.
- ▶ PAC bounds for dynamical systems.

ARMAX models

Recall ARX models

$$\mathbf{y}(t) = - \sum_{i=1}^{n_a} a_i \mathbf{y}(t-i) + \sum_{i=1}^{n_b} b_i \mathbf{u}(t-i) + \mathbf{e}(t)$$

ARX models can be viewed as

- ▶ Data generators
- ▶ If we forget $\mathbf{e}(t)$ as predictors:

$$\hat{\mathbf{y}}(t) = - \sum_{i=1}^{n_a} a_i \mathbf{y}(t-i) + \sum_{i=1}^{n_b} b_i \mathbf{u}(t-i)$$

$\hat{\mathbf{y}}(t)$ – predicted output based on past outputs and inputs.

Noise $\mathbf{e}(t)$ – prediction error.

Any learning algorithm which from data estimates a_i, b_i learns an

- ▶ an estimate of the noisy model which generates the data
- ▶ a good predictor.

ARX as state-space models

ARX as state-space model:

$$\mathbf{x}(t) = \begin{bmatrix} \mathbf{y}(t-1) & \mathbf{y}(t-2) & \cdots & \mathbf{y}(t-n_a) & \mathbf{u}(t-1) & \mathbf{u}(t-2) & \vdots & \mathbf{u}(t-n_b) \end{bmatrix}$$

$$\begin{aligned} \mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{u}(t) + K\mathbf{e}(t) \\ \mathbf{y}(t) &= C\mathbf{x}(t) + \mathbf{e}(t) \end{aligned}$$

But $\mathbf{e}(t) = \mathbf{y}(t) - C\mathbf{x}(t)$ and hence

$$\begin{aligned}\mathbf{x}(t+1) &= (A - KC)\mathbf{x}(t) + B\mathbf{u}(t) + K\mathbf{y}(t) \\ \mathbf{y}(t) &= C\mathbf{x}(t) + (\mathbf{y}(t) - C\mathbf{x}(t))\end{aligned}$$

Note that we obtained a state-space representation with inputs $\begin{bmatrix} \mathbf{u}(t) \\ \mathbf{y}(t) \end{bmatrix}$ and which generates the best possible predictors of $\mathbf{y}(t)$.

$$\begin{aligned}\mathbf{x}(t+1) &= (A - KC)\mathbf{x}(t) + \begin{bmatrix} B & K \end{bmatrix} \begin{bmatrix} \mathbf{u}(t) \\ \mathbf{y}(t) \end{bmatrix} \\ \hat{\mathbf{y}}(t) &= C\mathbf{x}\end{aligned}$$

Exercise: do the calculations for ARX

$$\mathbf{y}(t) = 0.8\mathbf{y}(t-1) - 0.12\mathbf{y}(t-2) + 0.5\mathbf{u}(t-1) + \mathbf{u}(t-2) + \mathbf{e}(t)$$

Note that $(A - KC)^{n+1} = 0$, i.e.,

$$\mathbf{x}(t) = \sum_{i=0}^n (A - KC)^i \begin{bmatrix} B & K \end{bmatrix} \begin{bmatrix} \mathbf{u}(t-i) \\ \mathbf{y}(t-i) \end{bmatrix}$$

ARMAX models

What if the data is generated as follows (ARMAX model)

$$\mathbf{y}(t) = \sum_{i=1}^{n_a} a_i \mathbf{y}(t-i) + \sum_{i=1}^{n_b} b_i \mathbf{u}(t-i) + \sum_{i=1}^{n_c} c_i \mathbf{e}(t-i) + \mathbf{e}(t)$$

$\mathbf{e}(t)$ – noise

Model contains a moving average part (MA) of the noise represented by the coefficients c_i .

Prediction:

$$\hat{\mathbf{y}}(t) = \sum_{i=1}^{n_a} a_i \mathbf{y}(t-i) + \sum_{i=1}^{n_b} b_i \mathbf{u}(t-i) + \sum_{i=1}^{n_c} c_i \hat{\mathbf{e}}(t-i)$$

$$\hat{\mathbf{e}}(t-i) = \mathbf{y}(t-i) - \hat{\mathbf{y}}(t-i)$$

ARMAX models as state-space models

$$\begin{aligned}\mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{u}(t) + K\mathbf{e}(t) \\ \mathbf{y}(t) &= C\mathbf{x}(t) + \mathbf{e}(t)\end{aligned}$$

with $\mathbf{e}(t) = \mathbf{y}(t) - C\mathbf{x}(t)$ and hence

$$\begin{aligned}\mathbf{x}(t+1) &= (A - KC)\mathbf{x}(t) + B\mathbf{u}(t) + K\mathbf{y}(t) \\ \mathbf{y}(t) &= C\mathbf{x}(t) + (\mathbf{y}(t) - C\mathbf{x}(t))\end{aligned}$$

Difference w.r.t. ARX models: $(A - KC)^{n+1} \neq 0$, i.e., is not nilpotent in general.

General stochastic state-space models

$$\begin{aligned}\mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{u}(t) + F\mathbf{v}(t) \\ \mathbf{y}(t) &= C\mathbf{x}(t) + H\mathbf{v}(t)\end{aligned}$$

State-space models of ARX, ARMA are special case.

$\mathbf{v}(t)$ and $H\mathbf{v}(t)$ are process and measurement noise, respectively, zero mean, i.i.d.

Represent modeling or measurement errors.

Question: how to interpret them as predictors ? Answer: Kalman-filters

Kalman Filters

$\hat{\mathbf{x}}(t \mid t_0)$ – the best linear estimate of $\mathbf{x}(t)$ given the measurements $\{\mathbf{y}(s), \mathbf{u}(s)\}_{s \leq t}$, orthogonal projection

$$\hat{\mathbf{x}}(t \mid t_0) = E^{\mathbf{p}(t|t_0)} \mathbf{x}(t)$$

$$\mathbf{p}(t) = [\mathbf{y}(t-1), \mathbf{u}(t-1) \quad \mathbf{y}(t-2), \mathbf{u}(t-2) \quad \cdots \mathbf{y}(t_0), \mathbf{u}(t_0)]^T$$

$\mathbf{p}(t \mid t_0)$ – past (possibly infinite) measurements and inputs between t_0 and $t-1$

Kalman-filter (Apollo): there exists gains K_t satisfying a recursion

$$K_{t+1} = f(K_t, A, C, H, F, E[\mathbf{v}\mathbf{v}^T])$$

$$\hat{\mathbf{x}}(t+1 \mid t_0) = A\hat{\mathbf{x}}(t \mid t_0) + B\mathbf{u}(t) + K_t\mathbf{e}(t \mid t_0)$$

$$\hat{\mathbf{y}}(t|t_0) = C\hat{\mathbf{x}}(t|t_0)$$

$$\mathbf{e}(t \mid t_0) = \mathbf{y}(t) - \hat{\mathbf{y}}(t|t_0)$$

Usage: filter out noise, $\hat{\mathbf{x}}(t \mid t_0)$ represents the best estimate of the state given the past measurements and inputs.

Stationary Kalman filter

If A is Schur (stable) + some technical conditions $\implies$
 $\hat{\mathbf{x}}(t) \triangleq \lim_{t_0 \rightarrow -\infty} \hat{\mathbf{x}}(t \mid t_0)$ – best prediction as the number of past measurements $t - t_0$ goes to infinity.

$K = \lim_{t \rightarrow \infty} K_t$ – stationary Kalman gain, $A - KC$ is Schur (stable).

Instead of letting $t \rightarrow \infty$ for a fixed t_0 we let t_0 go to $-\infty$.

Intuition: we observe the data after letting the system run for a sufficiently long time.

Stochastic systems started at $-\infty$

Noisy LTI stochastic system

$$\begin{aligned}\mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{u}(t) + F\mathbf{v}(t), \mathbf{x}(-\infty) = 0 \\ \mathbf{y}(t) &= C\mathbf{x}(t) + H\mathbf{v}(t)\end{aligned}$$

such that A is stable.

Unique wide-sense stationary solution $\mathbf{x}(t)$ and $\mathbf{y}(t)$, i.e., $E[\mathbf{x}(t)] = 0$, $E[\mathbf{y}(t)] = 0$, and their autocovariances depend only on the time difference $t - s$.

$$\begin{aligned}\mathbf{x}(t+1 \mid t_0) &= A\mathbf{x}(t \mid t_0) + B\mathbf{u}(t) + F\mathbf{v}(t), \mathbf{x}(t_0) = 0 \\ \mathbf{y}(t) &= C\mathbf{x}(t \mid t_0) + H\mathbf{v}(t)\end{aligned}$$

$$\mathbf{x}(t) = \lim_{t_0 \rightarrow -\infty} \mathbf{x}(t \mid t_0)$$

Innovation Form

Stable LTI system

$$\begin{aligned}\mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{u}(t) + F\mathbf{v}(t), \mathbf{x}(-\infty) = 0 \\ \mathbf{y}(t) &= C\mathbf{x}(t) + H\mathbf{v}(t)\end{aligned}$$

and it can be written in *innovation form*

$$\begin{aligned}\hat{\mathbf{x}}(t+1) &= A\hat{\mathbf{x}}(t) + B\mathbf{u}(t) + K\mathbf{e}(t) \\ \mathbf{y}(t) &= C\hat{\mathbf{x}}(t) + \mathbf{e}(t),\end{aligned}$$

where

$$\mathbf{e}(t) = \mathbf{y}(t) - E^{\mathbf{p}(t)}[\mathbf{y}(t)], \quad \hat{\mathbf{x}}(t) = E^{\mathbf{p}(t)}[\mathbf{x}(t)]$$

$$\mathbf{p}(t) = [\mathbf{u}(t), \mathbf{y}(t-1), \mathbf{u}(t-1) \quad \mathbf{y}(t-2), \mathbf{u}(t-2) \quad \cdots \mathbf{y}(-k), \mathbf{u}(-k) \quad \cdots]^T$$

$\mathbf{e}(t)$ – innovation (prediction error), $\hat{\mathbf{x}}(t)$ – best estimate of the state given past measurements and inputs, $\mathbf{p}(t)$ – infinite past measurements and inputs.

Uniqueness and minimality of the innovation form

Assume there is no feedback from $\mathbf{y}$ to $\mathbf{u}$, i.e., $\mathbf{u}(t)$ does not depend on $\{\mathbf{y}(s)\}_{s \leq t}$.

The LTI system in innovation form

$$\begin{aligned}\hat{\mathbf{x}}(t+1) &= A\hat{\mathbf{x}}(t) + B\mathbf{u}(t) + K\mathbf{e}(t) \\ \mathbf{y}(t) &= C\hat{\mathbf{x}}(t) + \mathbf{e}(t),\end{aligned}$$

is a minimal dimension model of $\mathbf{y}(t)$, if and only if $(C, A, [K \ B])$ is a minimal deterministic system ((C, A) is observable, $(A, [K \ B])$ is controllable).

All minimal LTI representations of $\mathbf{y}(t)$ are isomorphic: there exists an invertible linear transformation T

$$TAT^{-1} = A', \quad TB = B', \quad CT^{-1} = C'$$

Minimal innovation as predictors

Minimal innovation form can be used as predictors for the output $\mathbf{y}(t)$.

$$\begin{aligned}\hat{\mathbf{x}}(t+1) &= A\hat{\mathbf{x}}(t) + B\mathbf{u}(t) + K\mathbf{e}(t) \\ \mathbf{y}(t) &= C\hat{\mathbf{x}}(t) + \mathbf{e}(t),\end{aligned}$$

where $\hat{\mathbf{x}}(t)$ is the best estimate of the state given past measurements and inputs, and $\mathbf{e}(t)$ is the innovation (prediction error), and, if $A - KC$ is stable, then

$$\begin{aligned}\hat{\mathbf{x}}(t+1) &= (A - KC)\hat{\mathbf{x}}(t) + B\mathbf{u}(t) + K\mathbf{y}(t) \\ \hat{\mathbf{y}}(t) &= C\hat{\mathbf{x}}(t)\end{aligned}$$

Minimal innovation as predictors: consequences for learning

Assume data generator (true system) is in minimal innovation form.

Finding the true system (up to change of basis) $\iff$ finding best predictor.

If the learned model is minimal in innovation form $\implies$ it is isomorphic to the true model $\implies$ it has the same behavior as the true one for any **u**.

Minimal system

$$\begin{aligned}\mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{u}(t) + H\mathbf{v}(t) \\ \mathbf{y}(t) &= C\mathbf{x}(t) + F\mathbf{e}(t)\end{aligned}$$

is minimal innovation form, if $F = I$ identity, $A - HC$ is stable.

Time-series prediction as learning problem

Assume $(\mathbf{y}, \mathbf{u})$ is generated by a minimal innovation form system

$$\begin{aligned}\mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{u}(t) + K\mathbf{e}(t) \\ \mathbf{y}(t) &= C\mathbf{x}(t) + \mathbf{e}(t)\end{aligned}$$

Learning: from a finite sample $\{y(t), u(t)\}_{t=0}^{N-1}$ of $\mathbf{y}(t)$ and $\mathbf{u}(t)$ find matrices $\hat{A}, \hat{B}, \hat{C}, \hat{K}$, such that

$$\begin{aligned}\hat{\mathbf{x}}(t+1) &= (\hat{A} - \hat{K}\hat{C})\hat{\mathbf{x}}(t) + \hat{B}\mathbf{u}(t) + \hat{K}\mathbf{y}(t) \\ \hat{\mathbf{y}}(t) &= \hat{C}\hat{\mathbf{x}}(t)\end{aligned}$$

is such that variance of the prediction error $E(\|\mathbf{y}(t) - \hat{\mathbf{y}}(t)\|^2)$ is minimized.

Learning problem for dynamical systems

- ▶ $\mathbb{X}$ – input-space, $\mathbb{Y}$ output space.
- ▶ $\mathbf{x}(t) \in \mathbb{X}$ – input process, $\mathbf{y}(t) \in \mathbb{Y}$ – output process,

$$t \in \mathbb{T} = \begin{cases} \mathbb{Z} & \text{discrete-time} \\ \mathbb{R}_+ & \text{continuous-time} \end{cases}$$

▶ Hypotheses:

$\mathcal{H} \subseteq \{ \text{functions of the form } h : \bigcup_{t \in \mathbb{T}} (\mathbb{X} \times \mathbb{Y})^{[0,t)} \rightarrow \mathbb{Y} \}.$

$h(\{\mathbf{x}(s), \mathbf{y}(s)\}_{0 \leq s < t})$ – prediction of output $\mathbf{y}(t)$ based on past values of the inputs and outputs.

Example: dynamical systems

If past measurements of labels are available

$$\left\{ \begin{array}{l} \hat{\mathbf{s}}(t+1) \\ \frac{d}{dt} \hat{\mathbf{s}}(t) \end{array} \right\} = f(\hat{\mathbf{s}}(t), \mathbf{x}(t), \widehat{\mathbf{y}}(t)), \quad \hat{\mathbf{s}}(0) = \hat{\mathbf{s}}_0,$$

$$h(\{\mathbf{x}(s), \mathbf{y}(s)\}_{0 \leq s < t}) := g(\hat{\mathbf{s}}(t)).$$

Learning problem for dynamical systems

- **Loss function:** $\ell : \mathbb{Y} \times \mathbb{Y} \rightarrow [0, +\infty)$

$\ell(\mathbf{y}(t), h(\{\mathbf{x}(s), \mathbf{y}(s)\}_{0 \leq s < t}))$ difference between the output predicted by $h \in \mathcal{H}$ and true output.

$\ell(y, y') = (y - y')^2$ (ℓ_2 loss), $\ell(y, y') = |y - y'|$ (ℓ_1 loss).

- **True error** for a hypothesis h :
Fixed final time prediction error:

$$\mathcal{L}(h) = E[\ell(\mathbf{y}(T), h(\{\mathbf{x}(s), \mathbf{y}(s)\}_{0 \leq s < T}))]$$

Long-term prediction error:

$$\mathcal{L}(h) = \lim_{T \rightarrow \infty} E[\ell(\mathbf{y}(T), h(\{\mathbf{x}(s), \mathbf{y}(s)\}_{0 \leq s < T}))]$$

Learning problem

Learning problem: find $h_{\star} \in \mathcal{H}$ such that $\mathcal{L}(h_{\star})$ is small from

- ▶ **Single time series (discrete-time)** sample from $\{(\mathbf{x}(t), \mathbf{y}(t))\}_{0 \leq t < N}$
- ▶ **Multiple time-series (discrete- or continuous time)** sample from i.i.d variable $\{(\mathbf{x}_i(t), \mathbf{y}_i(t))\}_{t \in [0, T]}$ $i = 1, \dots, N$, having the same distribution as $(\mathbf{x}, \mathbf{y})$.

Solution:

Define the **empirical error** for hypothesis h :

$$\hat{\mathcal{L}}_N(h) = \begin{cases} \frac{1}{N} \sum_{t=0}^{N-1} \ell(\mathbf{y}(t), h(\{\mathbf{x}(s), \mathbf{y}(s)\}_{0 \leq s < t})) & \text{single time-series} \\ \frac{1}{N} \sum_{i=1}^N \ell(\mathbf{y}(T), h(\{\mathbf{x}_i(s), \mathbf{y}_i(s)\}_{0 \leq s < T})) & \text{multiple time-series} \end{cases}$$

Let $h_{\star}$ be such that $(\hat{\mathcal{L}}_N(h) + \text{regularization term})$ is small.

Question:

What can we say about the true error $\mathcal{L}(h_{\star})$?

Learning for LTI systems

Data generator: LTI in innovation form

$$\begin{aligned}\mathbf{x}_g(t+1) &= A_g \mathbf{x}(t) + K_g \mathbf{e}_g(t), \\ \begin{bmatrix} \mathbf{y}(t) \\ \mathbf{u}(t) \end{bmatrix} &= C_g \mathbf{x}_g(t) + \mathbf{e}_g(t)\end{aligned}$$

A_g – stable, $\mathbf{e}_g$ i.i.d. white noise.

If no feedback (Granger-causality) from $\mathbf{y}$ to $\mathbf{u}$, then LTI in innovation form:

$$\begin{aligned}\mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{u}(t) + K\mathbf{e}(t), \\ \mathbf{y}(t) &= C\mathbf{x}(t) + D\mathbf{u}(t) + \mathbf{e}(t)\end{aligned}$$

A, B, K, C, D are functions of A_g, K_g, C_g .

Learning problem: predictor

Predictors: stable LTI systems parametrised by $\theta \in \Theta$

$$\begin{aligned}\hat{\mathbf{x}}(t+1) &= A(\theta)\hat{\mathbf{x}}(t) + B(\theta)\mathbf{u}(t) + K(\theta)\mathbf{y}(t), \quad \hat{\mathbf{x}}(0) = \mathbf{0}, \\ \hat{\mathbf{y}}_\theta(t \mid 0) &= C(\theta)\hat{\mathbf{x}}(t) + D(\theta)\mathbf{u}(t).\end{aligned}$$

Learning problem: loss functions

Learning data:

sample $\{y(t), u(t)\}_{t=0}^N$ of the process $\{\mathbf{y}(t), \mathbf{u}(t)\}_{t=0}^N$.

Empirical error:

$$\hat{\mathcal{L}}_N(\theta) \triangleq \frac{1}{N} \sum_{i=0}^{N-1} \|\hat{\mathbf{y}}_{\theta}(i \mid 0) - \mathbf{y}(i)\|^2.$$

True error: assume infinite past of $\mathbf{y}$ and $\mathbf{u}$.

$$\begin{aligned} \hat{\mathbf{x}}(t+1) &= \hat{A}(\theta)\hat{\mathbf{x}}(t) + \hat{B}(\theta)\mathbf{u}(t) + \hat{K}(\theta)\mathbf{y}(t), \quad \hat{\mathbf{x}}(-\infty) = 0, \\ \hat{\mathbf{y}}_{\theta}(t) &= \hat{C}(\theta)\hat{\mathbf{x}}(t) + \hat{D}(\theta)\mathbf{u}(t). \end{aligned}$$

True error:

$$\mathcal{L}(\theta) = E[\|\hat{\mathbf{y}}_{\theta}(t) - \mathbf{y}(t)\|^2]$$

Learning problem: statement

Learning data:

sample $\{y(t), u(t)\}_{t=0}^N$ of the process $\{\mathbf{y}(t), \mathbf{u}(t)\}_{t=0}^N$.

Learning problem:

From data find θ s.t. the true error $\mathcal{L}(\theta)$ is minimal.

Application to parameter estimation

Parameter estimation: if θ_{true} is the parameter of the data generator,

$$(A(\theta_{true}), B(\theta_{true}), K(\theta_{true}), C(\theta_{true}), D(\theta_{true})) = (A, B, K, C, D)$$

then for θ_* the true error is minimal, i.e., $\inf_{\theta} \mathcal{L}(\theta) = \mathcal{L}(\theta_{true})$.

θ_{true} – true parameter of the data generator.

Suppose identifiability

distance in a suitable Riemannian manifold

$$\overbrace{d(\theta_{true}, \theta)} \leq C(\mathcal{L}(\theta) - \mathcal{L}(\theta_{true}))$$

Minimizing true loss $\mathcal{L}(\theta) \iff$ minimizing parameter estimation error.

Solving the learning problem

Solving the learning problem: minimise $\hat{\mathcal{L}}_N(\theta)$ instead of $\mathcal{L}(\theta)$.
minimize a cost function in

$$\lim_{N \rightarrow +\infty} \hat{\mathcal{L}}_N(\theta) = \mathcal{L}(\theta)$$

Minimizing prediction error is known as *Prediction Error Identification (PEM)*.

Gradient descent for prediction error minimization

Minimize the empirical loss

$$\hat{\mathcal{L}}_N(\theta) = \frac{1}{N} \sum_{t=0}^{N-1} \|\hat{\mathbf{y}}_{\theta}(t | 0) - \mathbf{y}(t)\|^2$$

using gradient descent:

$$\theta_{k+1} = \theta_k - \alpha_k \nabla_{\theta} \hat{\mathcal{L}}_N(\theta_k)$$

where $\alpha_k > 0$ is the step size (learning rate).

Gradient computation:

$$\frac{\partial \hat{\mathcal{L}}_N(\theta)}{\partial \theta} = \frac{2}{N} \sum_{t=0}^{N-1} \frac{\partial \hat{\mathbf{y}}_{\theta}(t | 0)}{\partial \theta}^T (\hat{\mathbf{y}}_{\theta}(t | 0) - \mathbf{y}(t)) = \frac{2}{N} \sum_{t=0}^{N-1} \frac{\partial \mathbf{e}_{\theta}(t | 0)}{\partial \theta}^T \mathbf{e}_{\theta}(t | 0)$$

$$\mathbf{e}_{\theta}(t | 0) = \mathbf{y}(t) - \hat{\mathbf{y}}_{\theta}(t | 0)$$

Convergence: if the loss function is convex or α_k is sufficiently small, gradient descent converges to a local minimum of $\hat{\mathcal{L}}_N(\theta)$.

Computing $\frac{\partial \mathbf{e}_\theta(t|0)}{\partial \theta}$

$$\hat{\mathbf{x}}(t+1) = \hat{A}(\theta)\hat{\mathbf{x}}(t) + \hat{B}(\theta)\mathbf{u}(t) + \hat{K}(\theta)\mathbf{y}(t)$$

$$\frac{\partial \hat{\mathbf{x}}(t+1)}{\partial \theta} = \frac{\partial \hat{A}(\theta)}{\partial \theta} \hat{\mathbf{x}}(t) + \hat{A}(\theta) \frac{\partial \hat{\mathbf{x}}(t)}{\partial \theta} + \frac{\partial \hat{B}(\theta)}{\partial \theta} \mathbf{u}(t) + \frac{\partial \hat{K}(\theta)}{\partial \theta} \mathbf{y}(t)$$

$$\frac{\partial \mathbf{e}_\theta(t|0)}{\partial \theta} = -\frac{\partial \hat{\mathbf{x}}(t)}{\partial \theta} \hat{C}(\theta) - \frac{\partial \hat{C}(\theta)}{\partial \theta} \hat{\mathbf{x}}(t)$$

Problem of parametrization

We want the parametrizations to be minimal and identifiable.

Recall: identifiability requires that

$$TA(\theta)T^{-1}, TB(\theta), TC(\theta), D(\theta)) = (A(\theta'), B(\theta'), C(\theta'), D(\theta')) \\ \implies \theta = \theta' \text{ and } T = I$$

Data-driven local coordinates: choose θ so that for all θ' close enough to θ the equation above holds.

Alternative: during gradient update project $\frac{\partial \hat{\mathcal{L}}_N(\theta)}{\partial \theta}$ onto complement of the tangent space of the manifold defined by the identifiability constraints

$$TA(\theta)T^{-1}, TB(\theta), TC(\theta), D(\theta)) = (A(\theta'), B(\theta'), C(\theta'), D(\theta')) \\ \implies \theta = \theta' \text{ and } T = I$$

Example of identifiable parametrization

Controllable Canonical Form (CCF)

For a single-input system with transfer function

$$H(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_{n-1} z^{-(n-1)}}{1 + a_1 z^{-1} + \dots + a_n z^{-n}}$$

the controllable canonical form is:

$$A = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_n \\ 1 & 0 & \cdots & 0 & -a_{n-1} \\ 0 & 1 & \cdots & 0 & -a_{n-2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$C = [b_{n-1} \quad b_{n-2} \quad \cdots \quad b_1 \quad b_0], \quad D = 0$$

Canonical parametrization:

$\theta = (a_1, \dots, a_n, b_0, \dots, b_{n-1})$ so that no zero-pole cancelation and all roots of

$$p(z) = 1 + a_1 z^{-1} + \dots + a_n z^{-n}$$

are inside the unit circle.

Exercise

Consider a stable stochastic LTI system in innovation form:

$$\begin{aligned}\mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{u}(t) + K\mathbf{e}(t) \\ \mathbf{y}(t) &= C\mathbf{x}(t) + D\mathbf{u}(t) + \mathbf{e}(t)\end{aligned}$$

with $A = \begin{bmatrix} 0.8 & 0.1 \\ 0 & 0.6 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$, $K = \begin{bmatrix} 0.2 \\ 0.1 \end{bmatrix}$, $C = [1 \quad 0.5]$, $D = 0$, and $\mathbf{e}(t) \sim \mathcal{N}(0, 0.1^2)$.

- ▶ Check stability: verify that all eigenvalues of A are inside the unit circle.
- ▶ Generate $N = 10000$ samples of data $\{y(t), u(t)\}$ from the system with white noise input $\mathbf{u}(t) \sim \mathcal{N}(0, 1)$.
- ▶ Use the implementation of PEM from the SLIPPY package or Matlab System Identification Toolbox to identify an LTI system from the simulated data.

- ▶ Apply subspace identification (N4SID algorithm) to the same data.
- ▶ Compare both methods:
 - ▶ Compare estimated parameters $\hat{A}_{\text{PEM}}, \hat{A}_{\text{N4SID}}$ with the true A .
 - ▶ Compute prediction errors and one-step-ahead predictions for both methods.
 - ▶ Find the state transformation matrix T such that $T \hat{A}_{\text{N4SID}} T^{-1} \approx \hat{A}_{\text{PEM}}$.

Asymptotic consistency of PEM for LTI systems

$\Theta_\star = \{\theta_\star \mid \min_{\theta \in \Theta} \mathcal{L}(\theta) = \mathcal{L}(\theta_\star)\}$ – minimizers of the true loss.

Infinite past empirical error:

$$V_N(\theta) = \frac{1}{N} \sum_{t=0}^{N-1} \|\hat{\mathbf{y}}_\theta(t) - \mathbf{y}(t)\|^2$$

$\hat{\mathbf{y}}_\theta(t) = \lim_{t_0 \rightarrow -\infty} \hat{\mathbf{y}}_\theta(t \mid t_0)$, i.e. the predictor θ was started infinitely far in the past:

$$\hat{\mathbf{x}}(t+1) = A(\theta)\hat{\mathbf{x}}(t) + B(\theta)\mathbf{u}(t) + K(\theta)\mathbf{y}(t),$$

$$\hat{\mathbf{x}}(-\infty) = 0$$

$$\hat{\mathbf{y}}_\theta(t) = C(\theta)\hat{\mathbf{x}}(t) + D(\theta)\mathbf{u}(t)$$

$$\lim_{t \rightarrow \infty} \hat{\mathbf{y}}_\theta(t) - \hat{\mathbf{y}}_\theta(t \mid 0) = 0 \quad \lim_{N \rightarrow \infty} V_N(\theta) - \hat{\mathcal{L}}_N(\theta) = 0.$$

For large N , $V_N(\theta) \approx \hat{\mathcal{L}}_N(\theta)$.

Asymptotic results for PEM (LTI)

If $\hat{\theta}_N = \operatorname{argmin}_{\theta \in \Theta} V_N(\theta)$, then under suitable conditions,
 $\hat{\theta}_N \rightrightarrows \Theta_\star$ as $N \rightarrow \infty$.

Minimizer of the the infinite past empirical loss $V_N(\theta)$ converges to the set of true minimizers $\Theta_\star$ of the true loss $\mathcal{L}(\theta)$ as $N \rightarrow \infty$.

PEM under realizability and identifiability

Assume θ_{true} corresponds to the data generator:

$$\mathbf{x}(t+1) = (A(\theta_{true}) + K(\theta_{true})C(\theta_{true}))\mathbf{x}(t) + B(\theta_{true})\mathbf{u}(t) + K(\theta_{true})\mathbf{e}(t)$$

$$\mathbf{x}(-\infty) = 0$$

$$\mathbf{y}(t) = C(\theta_{true})\mathbf{x}(t) + D(\theta_{true})\mathbf{u}(t) + \mathbf{e}(t)$$

is a stable LTI in innovation form.

Assume that Θ is identifiable, $\mathbf{u}$ is persistently exciting.

Then θ_{true} is the unique minimizer of $\mathcal{L}(\theta)$.

Under some mild assumptions:

- ▶ $\hat{\theta}_N \rightarrow \theta_{true}$ as $N \rightarrow \infty$
- ▶ The probability distribution of $\hat{\theta}_N$ converges to a Gaussian distribution centered at θ_{true} with variance $\frac{C}{\sqrt{N}}$ as $N \rightarrow \infty$ for a suitable N .

Generalisation bounds

Desired results:

$$P(\mathcal{L}(h_*) \leq \underbrace{\hat{\mathcal{L}}_N(h_*)}_{\text{known from data}} + \underbrace{r(N, \delta, \dots)}_{\text{functions of } N, \delta}) > 1 - \delta$$

and/or, oracle inequality:

$$P(\mathcal{L}(h_*) \leq \underbrace{\inf_h \mathcal{L}(h)}_{\text{best prediction}} + r(N, \delta, \dots)) > 1 - \delta.$$

Probability is taken over all random choices of training data.

$1 - \delta$ is the confidence level for the inequality, ideally $\delta = 0.02, 0.05$.

Typically, we want: $r(N, \delta, \dots) = C \frac{\ln(1/\delta)+1}{\sqrt{N}}$

Ways to derive it:

- ▶ PAC bounds.
- ▶ PAC-Bayesian bounds.
- ▶ Bounds specific to certain learning algorithms.

PAC (Probably Approximately Correct) bounds

$$P \left(\forall h \in \mathcal{H} : \mathcal{L}(h) - \hat{\mathcal{L}}_N(h) < C \frac{\ln(1/\delta) + 1}{\sqrt{N}} \right) > 1 - \delta$$

The bound holds for all h , thus for the learned model $h_\star$ too.

Results in an **oracle** inequality for $h_\star = \inf_h \hat{\mathcal{L}}_N(h)$:

$$P \left(\mathcal{L}(h_\star) \leq \inf_h \mathcal{L}(h) + \tilde{C}(C) \frac{\ln(1/\delta) + 1}{\sqrt{N}} \right) > 1 - \delta$$

Conceptually easy, but conservative.

C depends on the complexity of the model class, examples:
Vapnik-Chervonenkis dimension, Rademacher complexity.

E.g.: for RNNs learned from N i.i.d. time series of length T , PAC bounds are exponential in T .

Using PAC bounds for learning

1) Generalisation gap bound for any learning algorithm:

$$P \left(\forall h \in \mathcal{H} : \mathcal{L}(h) \leq \hat{\mathcal{L}}_N(h) + C \frac{\ln(1/\delta) + 1}{\sqrt{N}} \right) > 1 - \delta$$

This implies that for the learned model h_* :

$$P \left(\mathcal{L}(h_*) \leq \hat{\mathcal{L}}_N(h_*) + C \frac{\ln(1/\delta) + 1}{\sqrt{N}} \right) > 1 - \delta$$

2) Oracle bound for Empirical Risk Minimizer (ERM):

Let $h_* = \operatorname{argmin}_{h \in \mathcal{H}} \hat{\mathcal{L}}_N(h)$. Then:

$$P \left(\mathcal{L}(h_*) \leq \inf_{h \in \mathcal{H}} \mathcal{L}(h) + \tilde{C} \frac{\ln(2/\delta) + 1}{\sqrt{N}} \right) > 1 - \delta$$

- ▶ Oracle bounds tell us how far the performance (true error) of the learned model is from the best achievable true error.
- ▶ PAC bounds are relative: they tell the difference between empirical error and true error, or between the true error achieved by the learned model and the best possible true error. They do not say how big those errors are.

PAC bounds as finite-sample bounds

PAC bounds can be used to estimate the number of data points needed for performance.

For every $\epsilon > 0$ (error margin) we can choose a minimal number of data points N_ϵ such that for $N > N_\epsilon$:

$$P\left(\mathcal{L}(h_\star) \leq \hat{\mathcal{L}}_N(h_\star) + \epsilon\right) > 1 - \delta$$

$N_\epsilon = \left(\frac{\tilde{\mathcal{C}}(\ln(1/\delta)+1)}{\epsilon}\right)^2$ – minimal number of data points so that the true error is within ϵ distance from the empirical error.

If $h_\star = \operatorname{argmin}_{h \in \mathcal{H}} \hat{\mathcal{L}}_N(h)$:

$$P \left(\mathcal{L}(h_\star) \leq \inf_{h \in \mathcal{H}} \mathcal{L}(h) + \epsilon \right) > 1 - \delta$$

$$N_\epsilon = \left(\frac{\tilde{C}(\ln(2/\delta)+1)}{\epsilon} \right)^2$$

N_ϵ – minimal number of data points to be within ϵ margin from the best achievable true error.

PAC-Bayesian bound

for any **prior distribution** π on models, any $\lambda > 0$

$$P \left(\text{for any posterior } \rho : E_{h \sim \rho} \mathcal{L}(h) < E_{h \sim \rho} \hat{\mathcal{L}}_N(h) + \frac{1}{\lambda} \left[KL(\rho \mid \pi) + \ln\left(\frac{1}{\delta}\right) + \frac{\lambda C_\pi}{\sqrt{N}} \right] \right) > 1 - \delta$$

$KL(\rho \mid \pi)$ – KL divergence, ρ – **posterior distribution on models**

Getting generalisation bounds:

- ▶ choose a prior π , a data-dependent **posterior** distribution ρ_N s.t. $KL(\rho_N \mid \pi) < O(\sqrt{N})$.
- ▶ Learned model h_* : sample/mean/max. likelihood of ρ_N .
- ▶ Choose $\lambda = O(\sqrt{N})$

$$P \left(\mathcal{L}(h_*) \leq \hat{\mathcal{L}}_N(h_*) + \tilde{C} \frac{\ln(1/\delta) + 1}{\sqrt{N}} \right) > 1 - \delta.$$

Works for learning with regularisation (encoded by π).

Learning based on PAC-Bayesian error bounds

Learn by

1. find a posterior $\rho = \hat{\rho}_N$ and a parameter $\lambda = \lambda_N$ such that

$$E_{h \sim \rho}[\hat{\mathcal{L}}_N(h)] + \frac{1}{\lambda} \left[KL(\rho \mid \pi) + \ln\left(\frac{1}{\delta}\right) + \frac{\lambda C_\pi}{\sqrt{N}} \right]$$

is small.

2. $h_\star$ is one of the following:
 - ▶ $h_\star$ random sample from $\hat{\rho}_N$, or
 - ▶ most likely model, i.e. $h_\star = \sup_{h \in \mathcal{H}} \hat{\rho}_N(h)$, or
 - ▶ $h_\star$ is the mean model $E_{h \sim \hat{\rho}_N}[h]$.

Explicit formula for $\hat{\rho}_N$.

Analytic bound (with high probability) on the generalization gap

$$\mathcal{L}(h_\star) - \hat{\mathcal{L}}_N(h_\star)$$

Learning with PAC-Bayesian bounds and maximal likelihood

Explicit formula for $\hat{\rho}_N$ (Gibbs posterior)

$$\hat{\rho}_N(h) = \frac{1}{Z} e^{-\lambda \hat{\mathcal{L}}_N(h)} \pi(h), \quad \hat{\rho}_N = \operatorname{argmin}_{\rho} \left(E_{h \sim \rho} \hat{\mathcal{L}}_N(h) + KL(\rho \| \pi) \right)$$

Maximum likelihood and prediction error:

$$\hat{h} = \operatorname{argmax}_{h \in \mathcal{H}} \hat{\rho}_N(h) = \operatorname{argmin}_{h \in \mathcal{H}} \left(\hat{\mathcal{L}}_N(h) - \frac{1}{\lambda} \ln(\pi(h)) \right)$$

Using prior π introduces assumption on the model structure, it can be viewed as a regularisation

Oracle inequality

$$P \left(E_{h \sim \hat{\rho}_N} \mathcal{L}(h) < \inf_{\rho} \left\{ E_{h \sim \rho} \hat{\mathcal{L}}_N(h) + \frac{2}{\lambda} \left[KL(\rho \| \pi) + \ln\left(\frac{1}{\delta}\right) + \frac{\lambda C_{\pi}}{\sqrt{N}} \right] \right\} \right) > 1 - \delta$$

From PAC-Bayesian bounds to PAC bounds

Models sampled randomly from posterior With a probability at least $(1 - 2\delta)$ over the data and over all samples h_* drawn from $\hat{\rho}_N$,

$$\mathcal{L}(h_*) \leq \hat{\mathcal{L}}_N(h_*) + \frac{\ln \frac{\hat{\rho}_N(\theta_*)}{\pi(\theta_*)} + \ln \frac{1}{\delta} + C_\pi}{\sqrt{N}}$$

PAC bounds for uniform/Gaussian priors With probability at least $1 - 2\delta$ over data,

$$\mathcal{L}(\hat{h}) \leq \hat{\mathcal{L}}_N(\hat{h}) + \frac{2\sigma^2 L}{\sqrt{N}} + \frac{C(\hat{h}) + \ln \frac{1}{\delta} + C_\pi}{\sqrt{N}}$$

C_π - analytic formula

Stable dynamical systems as data generators and models

Data generator: Lyapunov stable systems

$$\mathbf{s}_g(t+1) = f_g(\mathbf{s}_g(t), \mathbf{e}_g(t)), \quad \mathbf{e}_g - \text{bounded i.i.d. noise.}$$

$$\begin{bmatrix} \mathbf{y}(t) \\ \mathbf{u}(t) \end{bmatrix} = h_g(\mathbf{s}_g(t), \mathbf{e}_g(t))$$

Stability: small past perturbations in the past do not lead to large errors in the future.

Hypothesis: stable dynamical systems parametrized by θ

$$\begin{aligned} \hat{\mathbf{s}}(t+1) &= f_\theta(\hat{\mathbf{s}}(t), \mathbf{u}(t), \mathbf{y}(t)), \quad \hat{\mathbf{s}}(0) = \hat{\mathbf{s}}_0, \\ \hat{\mathbf{y}}(t \mid 0) &:= h(\{\mathbf{u}(s), \mathbf{y}(s)\}_{0 \leq s < t}) := g_\theta(\hat{\mathbf{s}}(t)). \end{aligned}$$

Data generator and predictors can be RNNs.

Learning problem for a single time-series

Learning data:

sample $\{x(t), y(t)\}_{t=0}^N$ of the process $\{\mathbf{u}(t), \mathbf{y}(t)\}_{t=0}^N$.

Empirical loss:

$$\hat{\mathcal{L}}_N(\theta) \triangleq \frac{1}{N} \sum_{i=0}^{N-1} \ell(\mathbf{y}(t), h(\{\mathbf{u}(s), \mathbf{y}(s)\}_{0 \leq s < t})).$$

True error: long-term prediction error

$$\mathcal{L}(h) = \lim_{T \rightarrow \infty} E[\ell(\mathbf{y}(T), h(\{\mathbf{u}(s), \mathbf{y}(s)\}_{0 \leq s < T}))]$$

Long-term prediction error $\iff$ prediction error in stationary regime

Data generator stable: $\mathbf{u}, \mathbf{y}$ stationary, ergodic, weakly dependent.

Stationary regime of the predictor (using infinite past):

$$\begin{aligned}\hat{\mathbf{s}}(t+1) &= f_{\theta}(\hat{\mathbf{s}}(t), \mathbf{u}(t), \mathbf{y}(t)), \quad \hat{\mathbf{s}}(-\infty) = 0, \\ \hat{\mathbf{y}}_{\theta}(t) &= h_{\theta}(\hat{\mathbf{s}}(t), \mathbf{u}(t)).\end{aligned}$$

Due to stability $\hat{\mathbf{s}}(t)$ is stationary, and long-term prediction error is the same as final-time one:

$$\mathcal{L}(\theta) = E[\ell(\hat{\mathbf{y}}_{\theta}(t), \mathbf{y}(t))]$$

Example: RNNs

data generators is an RNN with $\|A_g\|_2 < 1$

$$\begin{aligned}\mathbf{s}_g(t+1) &= \text{ReLu}(A_g \mathbf{s}_g(t) + K_g \mathbf{e}_g(t) + B_g \mathbf{u}(t)) \\ \begin{bmatrix} \mathbf{y}(t) \\ \mathbf{u}(t) \end{bmatrix} &= \tanh(C_g \mathbf{s}_g(t) + F_g \mathbf{e}_g(t))\end{aligned}$$

hypotheses are RNNs with parameters (weights) $\theta = (A, B, K, C)$
 $\|A\|_2 < 1$

$$\begin{aligned}\hat{\mathbf{s}}(t+1) &= \text{ReLu}(A\hat{\mathbf{s}}(t) + B\mathbf{u}(t) + K\mathbf{y}(t)), \quad \hat{\mathbf{s}}(0) = 0 \\ h(\{\mathbf{u}(s), \mathbf{y}(s)\}_{0 \leq s < t}) &= \tanh(C\hat{\mathbf{s}}(t))\end{aligned}$$

Example: linear systems

Data generator: stable linear system

$$\begin{aligned}\mathbf{s}_g(t+1) &= A_g \mathbf{s}_g(t) + K_g \mathbf{e}_g(t), \\ \begin{bmatrix} \mathbf{y}(t) \\ \mathbf{u}(t) \end{bmatrix} &= C_g \mathbf{s}_g(t) + \mathbf{e}_g(t)\end{aligned}$$

$\|A_g\|_2 < 1$ and $\mathbf{e}_g$ i.i.d. white noise.

Predictors: stable linear systems parametrised by $\theta \in \Theta$ ($\hat{A}(\theta)$ has all its eigenvalues inside the unit circle)

$$\begin{aligned}\hat{\mathbf{s}}(t+1) &= \hat{A}(\theta)\hat{\mathbf{s}}(t) + \hat{B}(\theta)\mathbf{u}(t) + \hat{K}(\theta)\mathbf{y}(t), \quad \hat{\mathbf{s}}(0) = 0, \\ h(\{\mathbf{u}(s), \mathbf{y}(s)\}_{0 \leq s < t}) &= \hat{C}(\theta)\hat{\mathbf{s}}(t).\end{aligned}$$

Example: for linear systems data generator is the best predictor

If no feedback (Granger-causality) from $\mathbf{y}$ to $\mathbf{u}$, then the data generator is a noisy input-output model:

$$\begin{aligned}\mathbf{s}(t+1) &= A\mathbf{s}(t) + B\mathbf{u}(t) + K\mathbf{e}(t), \\ \mathbf{y}(t) &= C\mathbf{s}(t) + \mathbf{e}(t)\end{aligned}$$

A, B, K, C, D are one-to-one functions of A_g, K_g, C_g and the predictor

$$\begin{aligned}\hat{\mathbf{s}}(t+1) &= (A - KC)\hat{\mathbf{s}}(t) + B\mathbf{u}(t) + K\mathbf{y}(t), \quad \hat{\mathbf{u}}(0) = 0, \\ h(\{\mathbf{u}(s), \mathbf{y}(s)\}_{0 \leq s < t}) &= C\hat{\mathbf{s}}(t).\end{aligned}$$

gives the minimal true loss.

Learning data generator $\iff$ finding best predictor

PAC-Bayesian bounds for non-linear systems

Theorem 1 (Eringis & et al. AAAI 2024)

For all $\lambda > 0$, $\delta \in [0, 0.5)$, for all priors π

$$P \left(\forall \rho : E_{h \sim \rho} \mathcal{L}(h) \leq E_{h \sim \rho} \hat{\mathcal{L}}_N(h) + r_N(\lambda, \pi, \rho, \delta) \right) > 1 - 2\delta$$

$$r_N(\lambda, \pi, \rho, \delta) \triangleq \frac{1}{\lambda} \left[KL(\rho || \pi) + \ln \frac{1}{\delta} + \frac{\lambda}{\sqrt{N}} C_\pi \right]$$

where C_π depends on the prior π and on the class of predictors.

Bounds can be computed. Extensions to unbounded linear RNNs (Eringis & et al, ICML 2024).